LM Physics of Climate

Anjali A. A. Piette

9 Radiative equilibrium with stellar absorption

9.1 Absorption of stellar radiation

Until now, we have used the two-stream approximation under the assumption that the atmosphere is transparent to incoming stellar radiation. In that case, all stellar radiation is absorbed by the surface, which re-radiates the energy in the infrared. However, realistic atmospheres do absorb at least some stellar radiation.

We can account for the absorption of stellar radiation in the atmosphere by modifying the equations we have seen previously. One way to see these modifications is that we are adding a third 'stream' to our existing upwards and downwards infrared streams, i.e., the optical stellar flux travelling downwards.

Before, we stated that in radiative equilibrium the heating rate, $-\frac{\mathrm{d}}{\mathrm{d}\overline{\tau}}(F_+ - F_-)$, is equal to zero. With our third stream, this is no longer the case, as the absorption of the infrared fluxes must be balanced alongside absorption of the stellar flux, F_s. Instead, radiative equilibrium can be expressed as:

$$-\frac{\mathrm{d}}{\mathrm{d}\overline{\tau}}(F_+ - F_-) + \frac{\mathrm{d}F_s}{\mathrm{d}\overline{\tau}} = 0. \tag{9.1}$$

Both F_- and F_s are travelling downwards, so are positive terms, while the F_+ term is negative as it travels upwards. Integrating equation 9.1 gives $F_+ - F_- - F_s =$ constant. At the top of the atmosphere, $F_+ = OLR$, $F_- = 0$ and $F_s = \sigma_{SB} T_{eq}^4 = OLR$, so the integration constant is zero. Therefore,

$$(F_+ - F_-) - F_s = 0. \tag{9.2}$$

The aim now is to find the temperature profile of the atmosphere, as we did before in the case of no stellar absorption. Recall that from Equation 8.7, we have $\frac{\mathrm{d}}{\mathrm{d}\overline{\tau}}(F_+ + F_-) = -(F_+ - F_-)$. Integrating this, and substituting 9.2, we get:

$$\begin{aligned}
[(F_+ + F_-)']_{F_+ + F_-}^{OLR} &= -\int_{\overline{\tau}}^{\overline{\tau}_\infty} F_s(\overline{\tau}')\mathrm{d}\overline{\tau}' \\
F_+ + F_- &= \int_{\overline{\tau}}^{\overline{\tau}_\infty} F_s(\overline{\tau}')\mathrm{d}\overline{\tau}' + OLR
\end{aligned} \tag{9.3}$$

In equation 8.6, we found an alternative way to write $\frac{\mathrm{d}}{\mathrm{d}\overline{\tau}}(F_+ - F_-)$ by directly substituting in the two-stream equations ($\frac{\mathrm{d}}{\mathrm{d}\overline{\tau}}(F_+ - F_-) = -(F_+ + F_-) + 2\sigma_{SB} T(\overline{\tau})^4$). We can do something similar here, but instead of setting $\frac{\mathrm{d}}{\mathrm{d}\overline{\tau}}(F_+ - F_-) = 0$, we set $\frac{\mathrm{d}}{\mathrm{d}\overline{\tau}}(F_+ - F_-) = \frac{\mathrm{d}F_s}{\mathrm{d}\overline{\tau}}$, as in Equation 9.1:

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}\overline{\tau}}(F_+ - F_-) = -(F_+ + F_-) + 2\sigma_{SB} T(\overline{\tau})^4 &= \frac{\mathrm{d}F_s}{\mathrm{d}\overline{\tau}} \\
2\sigma_{SB} T(\overline{\tau})^4 &= \frac{\mathrm{d}F_s}{\mathrm{d}\overline{\tau}} + \int_{\overline{\tau}}^{\overline{\tau}_\infty} F_s(\overline{\tau}')\mathrm{d}\overline{\tau}' + OLR,
\end{aligned} \tag{9.4}$$

where in the last line we have substituted Equation 9.3. The case without absorption of stellar radiation is recovered if we set $F_s = \text{constant} = OLR$, in which case $2\sigma_{SB}T(\overline{\tau})^4 = (1 + \overline{\tau}_\infty - \overline{\tau})OLR$, i.e. Equation 8.9.

Taking the derivative of equation 9.4 with respect to $\overline{\tau}$, we get:

$$\frac{\mathrm{d}}{\mathrm{d}\overline{\tau}}\left(2\sigma_{SB}T(\overline{\tau})^4\right) = \frac{\mathrm{d}^2F_s}{\mathrm{d}\overline{\tau}^2} - F_s. \tag{9.5}$$

The sign of the right-hand side of this equation tells us if temperature increases or decreases with height. It is useful here to define the local stellar flux decay rate,

$$\text{local stellar flux decay rate} = \left|\frac{1}{F_s}\frac{\mathrm{d}^2F_s}{\mathrm{d}\overline{\tau}^2}\right|. \tag{9.6}$$

This is the decay of stellar radiation with respect to the *infrared* optical depth, $\overline{\tau}$. This rate is higher when a layer of the atmosphere strongly absorbs radiation from the star. Since the Sun emits primarily in the UV/optical/near-infrared, absorbers of these wavelengths can increase the rate of solar flux decay. For example, ozone is a strong UV absorber in the Earth's atmosphere.

When the decay rate is less than one, $\frac{\mathrm{d}T}{\mathrm{d}\overline{\tau}} < 0$, and temperature decreases with height. However, when the stellar decay rate is greater than one, $\frac{\mathrm{d}T}{\mathrm{d}\overline{\tau}} > 0$ and temperature increases with height, as it does in the Earth's stratosphere. Another term for a region where temperature increases with height is a *thermal inversion*.

It is important to note that what determines the sign of the temperature gradient is the stellar flux decay *relative to the infrared optical depth*. So, temperature can increase with height when there is more absorption of stellar radiation, but also when infrared absorption is decreased.

9.2 Example: exponential decay of stellar flux

Let's illustrate the concepts above with an example. Consider an exponential decay of stellar radiation as it travels through an atmosphere: $F_s = OLRe^{-(\overline{\tau}_\infty - \overline{\tau})/\overline{\tau}_s}$. In this case, equation 9.4 becomes:

$$\begin{aligned}
\frac{2\sigma_{SB}T(\overline{\tau})^4}{OLR} &= 1 + \frac{1}{\overline{\tau}_s}e^{-(\overline{\tau}_\infty - \overline{\tau})/\overline{\tau}_s} + \int_{\overline{\tau}}^{\overline{\tau}_\infty} e^{-(\overline{\tau}_\infty - \overline{\tau}')/\overline{\tau}_s}\mathrm{d}\overline{\tau}' \\
&= 1 + \frac{1}{\overline{\tau}_s}e^{-(\overline{\tau}_\infty - \overline{\tau})/\overline{\tau}_s} + \overline{\tau}_s\left[e^{-(\overline{\tau}_\infty - \overline{\tau})/\overline{\tau}_s}\right]_{\overline{\tau}}^{\overline{\tau}_\infty} \\
&= 1 + \frac{1}{\overline{\tau}_s}e^{-(\overline{\tau}_\infty - \overline{\tau})/\overline{\tau}_s} + \overline{\tau}_s(1 - e^{-(\overline{\tau}_\infty - \overline{\tau})/\overline{\tau}_s}) \\
&= 1 + \overline{\tau}_s + \left(\frac{1}{\overline{\tau}_s} - \overline{\tau}_s\right)e^{-(\overline{\tau}_\infty - \overline{\tau})/\overline{\tau}_s}.
\end{aligned} \tag{9.7}$$

Temperature increases with height for $\overline{\tau}_s < 1$, decreases with height for $\overline{\tau}_s > 1$, and is constant for $\overline{\tau}_s = 1$ (see Figures 1 - 3). At the top of the atmosphere, the temperature is

$$T(\overline{\tau}_\infty) = (1 + 1/\overline{\tau}_s)^{1/4}T_{skin}, \tag{9.8}$$

where the skin temperature, $T_{skin} = 2^{-1/4}T_{eq}$, is the top-of-atmosphere temperature in the case with no absorption of stellar radiation (see Section 8). This expression reduces to the skin temperature when $\overline{\tau}_s$ is large. If $\overline{\tau}_s$ is small (i.e. the stellar radiation is rapidly absorbed), $T(\overline{\tau}_\infty)$ is much greater than the skin temperature.

If $\overline{\tau}_s$ is small and $\overline{\tau}_s \ll \overline{\tau}_\infty$ (i.e. the stellar radiation is absorbed high in the atmosphere), then in the deep atmosphere (where $\overline{\tau}$ is small), the exponential term in

equation 9.7 vanishes, leaving $\frac{2\sigma_{SB}T(\overline{\tau})^4}{OLR} = 1+\overline{\tau}_s$. Using the fact that $OLR = \sigma_{SB}T_{eq}^4 = 2\sigma_{SB}T_{skin}^4$, this means that in the deep atmosphere, $T = (1+\overline{\tau}_s)^{1/4}T_{skin} =$ constant. Thus, when stellar radiation is absorbed high in the atmosphere, the upper atmosphere becomes hotter than the skin temperature, and the deep atmosphere forms an isotherm that is close to the skin temperature (since $\overline{\tau}_s$ is small, see Figure 1). Note that, since the skin temperature is lower than the equilibrium temperature, the deep atmosphere is actually colder than it would have been if there were no atmosphere at all. This is known as the *anti-greenhouse effect*. It is also the mechanism behind the phenomenon of a 'nuclear winter', i.e. where explosions and fires loft dust and soot to high altitudes, where they efficiently absorb solar radiation and cool the surface significantly.

What if $\overline{\tau}_s$ is not small, but we still have $\overline{\tau}_s \ll \overline{\tau}_\infty$? This applies to giant planets, where the atmosphere is very deep and hence $\overline{\tau}_\infty$ can be much larger than $\overline{\tau}_s$, even if $\overline{\tau}_s$ is large. In this case, the deep atmosphere is still an isotherm at $T = (1+\overline{\tau}_s)^{1/4}T_{skin}$, but this temperature is higher than T_{skin}. In this scenario, the upper atmosphere equilibrates at the skin temperature, while the deep atmosphere exhibits greenhouse warming (Figure 4).

On Earth, we have a stratosphere where temperature increases with height, but the troposphere beneath it is *not* isothermal as we just calculated. This is because in reality, stellar radiation spans a range of wavelengths, and the atmosphere can absorb some but not all of these. On Earth, it is ozone which causes temperature to increase with height in the stratosphere. Ozone absorbs UV radiation from the Sun, but optical radiation from the Sun still makes it to the surface, which heats up. The warm surface then heats the troposphere from below - like heating a pan of water on a hob, this causes convection in the troposphere.

Equation 9.7 explains why, as the concentration of CO_2 in the Earth's atmosphere increases, the troposphere warms up but the stratosphere cools down. Increasing the concentration of CO_2 is equivalent to increasing the infrared total optical depth of the atmosphere, i.e. increasing $\overline{\tau}_\infty$. This reduces the temperature in the stratosphere, as shown in Figure 1

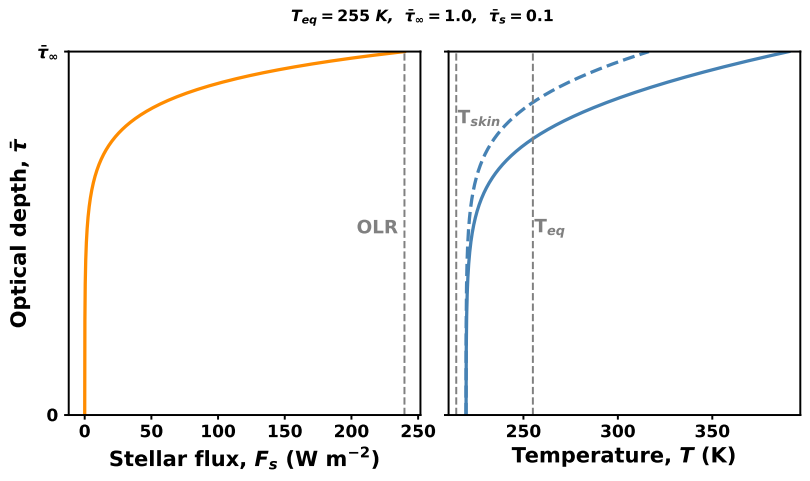


Figure 1: Stellar flux, F_s, and atmospheric temperature, T, as a function of $\overline{\tau}$ for the example of an exponentially decaying stellar flux, $F_s = OLR e^{-(\overline{\tau}_\infty - \overline{\tau})/\overline{\tau}_s}$. In this figure, $\overline{\tau}_\infty = 1.0$ and $\overline{\tau}_s = 0.1$ for the solid profiles. This means that the stellar flux is rapidly absorbed at the top of the atmosphere, causing a thermal inversion. For the dashed temperature profile, $\overline{\tau}_\infty = 1.1$, equivalent to increasing the concentration of greenhouse gases. This causes the stratosphere to cool compared to the solid line.

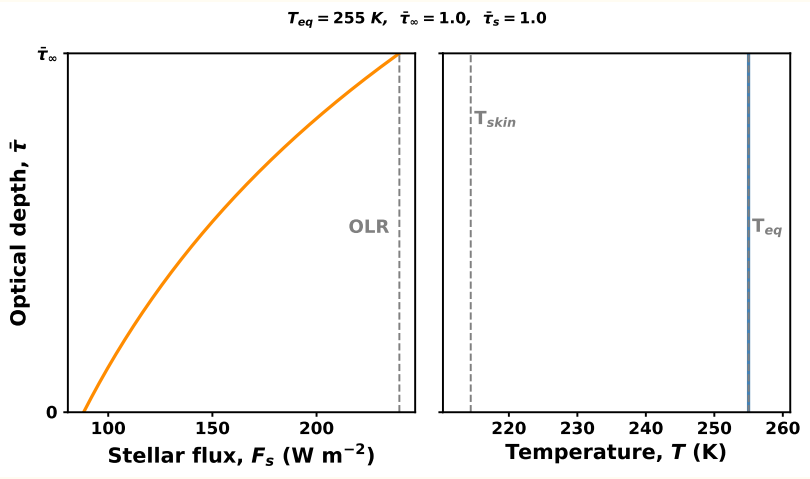


Figure 2: Stellar flux, F_s, and atmospheric temperature, T, as a function of $\overline{\tau}$ for the example of an exponentially decaying stellar flux, $F_s = OLR e^{-(\overline{\tau}_\infty - \overline{\tau})/\overline{\tau}_s}$. In this figure, $\overline{\tau}_\infty = 1.0$ and $\overline{\tau}_s = 1.0$. From Equation 9.7, this means that the temperature profile is isothermal at the equilibrium temperature.

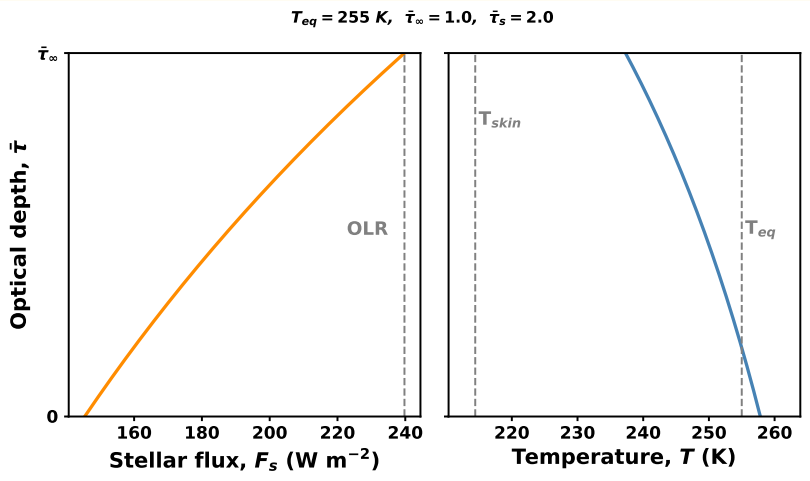


Figure 3: Stellar flux, F_s, and atmospheric temperature, T, as a function of $\overline{\tau}$ for the example of an exponentially decaying stellar flux, $F_s = OLRe^{-(\overline{\tau}_\infty - \overline{\tau})/\overline{\tau}_s}$. In this figure, $\overline{\tau}_\infty = 1.0$ and $\overline{\tau}_s = 2.0$. This means that the stellar flux is slowly absorbed through the atmosphere, and temperature decreases with height.

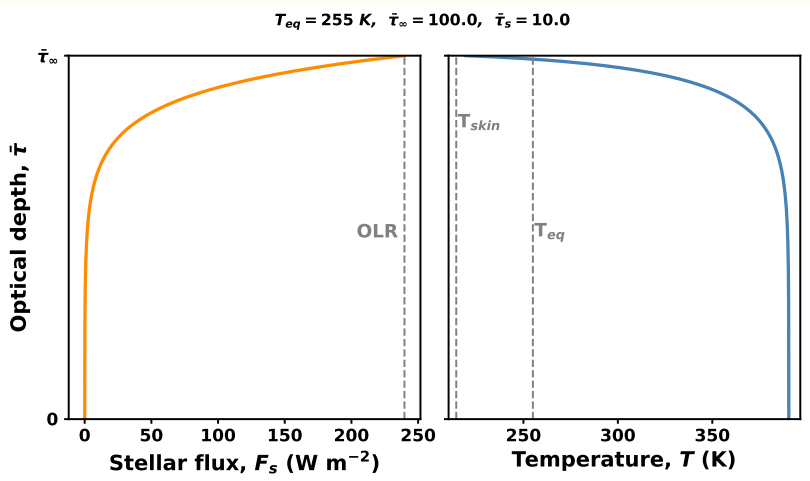


Figure 4: Stellar flux, F_s, and atmospheric temperature, T, as a function of $\overline{\tau}$ for the example of an exponentially decaying stellar flux, $F_s = OLRe^{-(\overline{\tau}_\infty - \overline{\tau})/\overline{\tau}_s}$. In this figure, $\overline{\tau}_\infty = 100.0$ and $\overline{\tau}_s = 10.0$. This means that the atmosphere is very optically thick (e.g. in the case of a giant planet), and although $\overline{\tau}_s$ is large, it is still much smaller than $\overline{\tau}_\infty$. As such, the deep atmosphere approaches an isotherm which is hotter than the skin temperature.